

INT202 Complexity of Algorithms

1. In the RSA method, suppose that $p = 5$, $q = 17$, and $e = 13$. First find the private key d corresponding to these parameters. Then decrypt the ciphertext messages, C , below to find the original (plaintext) messages.
 - a. $C = 12$
 - b. $C = 9$

2. Alice and Bob are using the RSA algorithm to communicate. Bob's public key is $e = 3$ and $n = 187$.
 - a. What is Bob's secret key?
 - b. Alice wants to send the message M to Bob. Bob receives 9. What was the message M sent by Alice?